

Critique and Discussion of Slide Figures

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1 First Chart

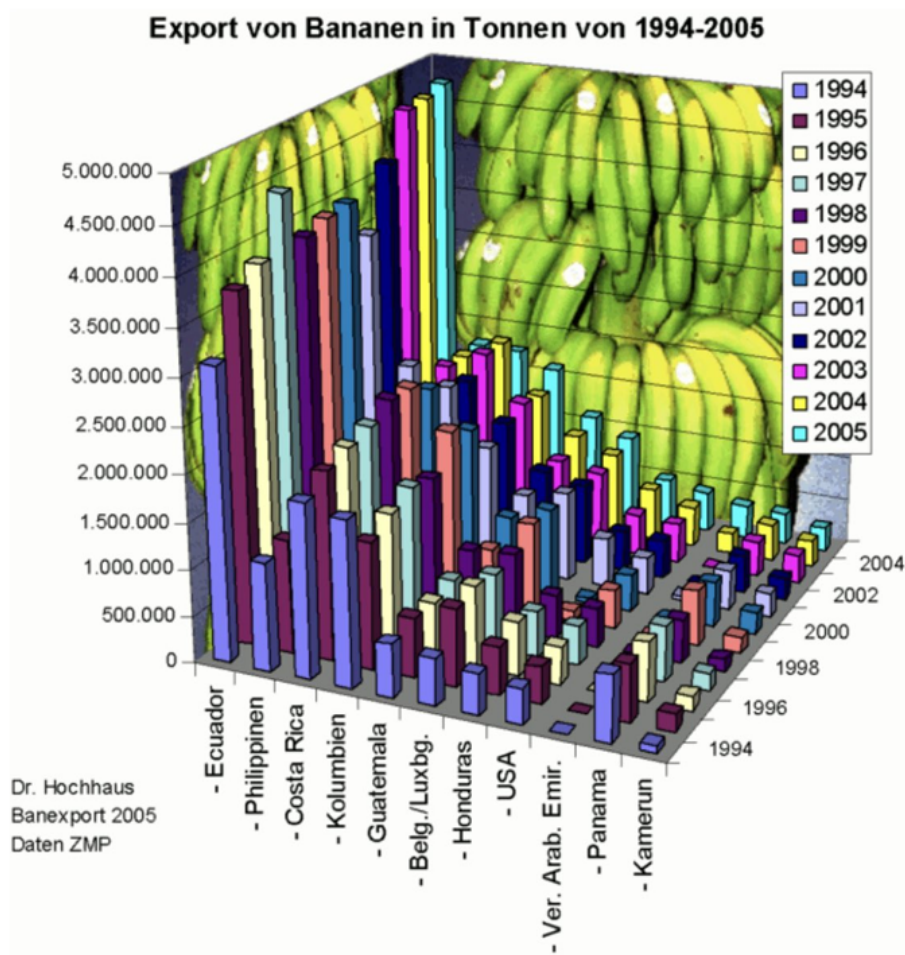


Figure 1: Page 2 of Data Visualization Slides

What the Chart Is About

The chart depicts the export of bananas (in tons) from various countries for the years 1994-2005.

Critique Based on the Guidelines

Chart Type (Data, I)

The 3D bar chart is not an effective choice. It makes it difficult to accurately compare the export volumes.

Readability (Graphical Objects, II)

The 3D perspective, overlapping bars, and busy background (bananas) significantly decrease readability, making it hard to discern the value of any individual bar.

Curve Comparison (Information, IV)

Comparing the height of the bars (data) is difficult due to the 3D effect. Comparing trends over time for each country is especially difficult.

Number of Curves (Information, IV)

The large number of bars (years and countries) contributes to visual clutter.

Informative Graphic (Context, V)

The chart does not convey as much information as other chart types would, for example, a grouped bar chart or a line chart.

Elegance (VI)

The overall design is not elegant; the 3D effect and the background image detract from the clarity of the data.

Gridlines (Graphical Objects, II)

There are gridlines (good), but the overall visual complexity overshadows the gridlines' benefit.

Annotations (Annotations, III)

The legend is there, but the order of the data is not by any logical ordering.

Alternative Proposed Chart

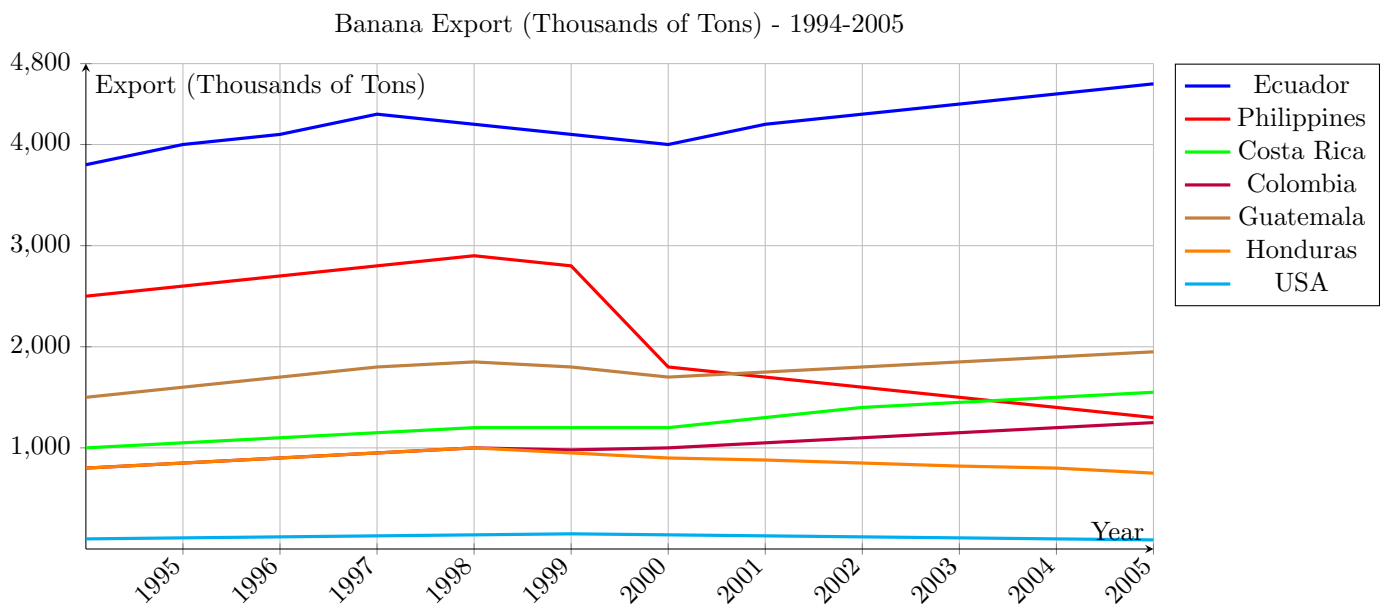


Figure 2: Banana Export Trends for Selected Countries (1994-2005). This line chart illustrates the export volumes of bananas (in thousands of tons) from 1994 to 2005 for several major producing countries. Key observations include Ecuador's consistently high export volume, a decline in exports from the Philippines, and steady growth from Costa Rica and Colombia. The data shows the evolution of the countries in this sector.

2 Second Chart

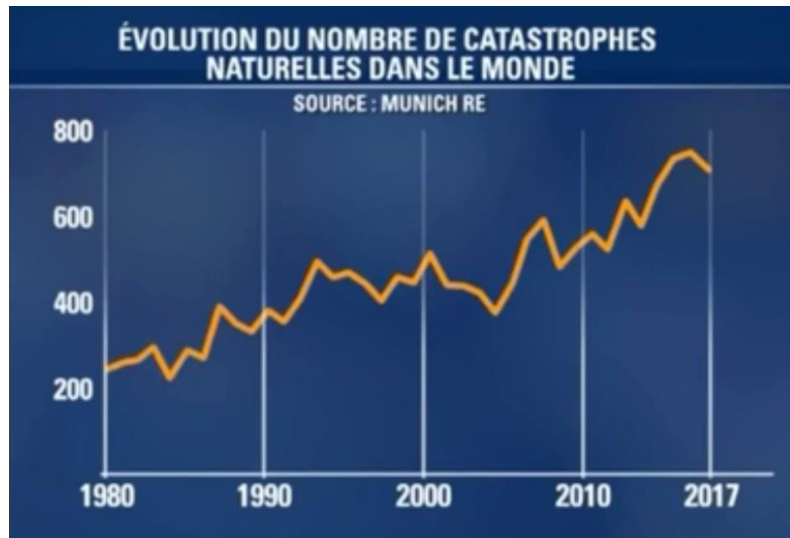


Figure 3: Page 30 of Data Visualization Slides

Critique Based on the Guidelines

- (III.1) The y-axis has a label but lacks the unit.
- (III.3) The graph's y-axis lacks a label indicating what the numbers represent (e.g., "Number of Disasters" or whatever unit Munich RE uses). **Improvement:** Adding this label is essential for clarity.
- (IV.6) The graph does not have averages or error bars.
- (IV.4) The curve can be removed, and the graph will still contain information.
- (VI.1) The graph has no legend, since it only includes one line.

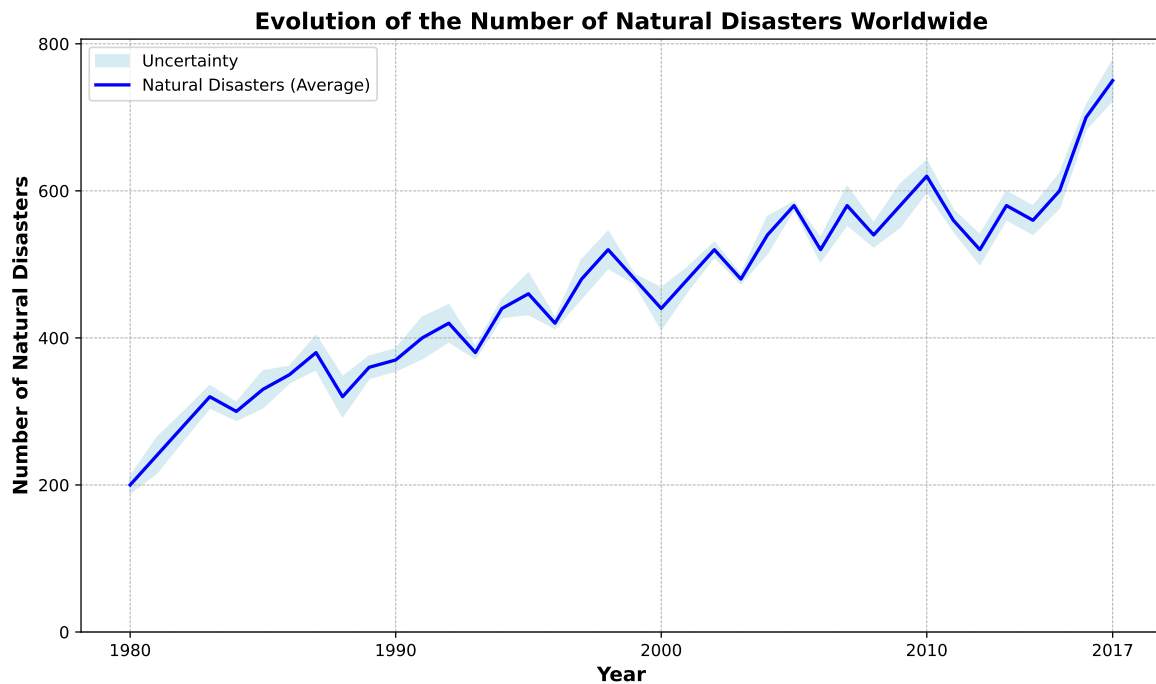


Figure 4: Evolution of the number of natural disasters worldwide from 1980 to 2017, with shaded region representing uncertainty. Source: Hypothetical Data.

3 Second Chart

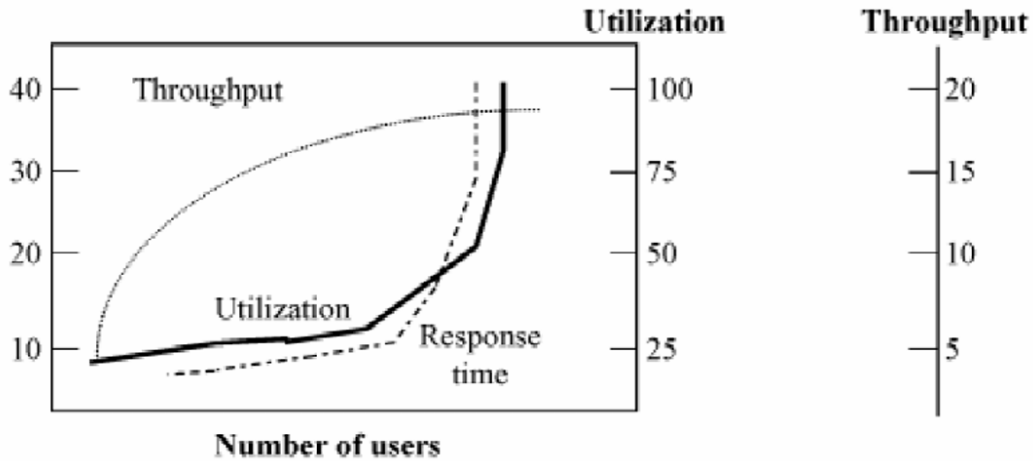


Figure 5: Page 10 of Data Visualization Slides

Critique Based on the Guidelines

This is just a sample!

- (I.1) **No Confidence Intervals:** The absence of confidence intervals for the Response time, Utilization, and Throughput graphs (if relevant) is a missed opportunity.
- (II.1) **Color:** The lines are black, it can be better.
- (II.2) **Curve Ambiguity:** The overlapping of the Throughput and Response time curves could be confusing. Consider alternative ways to display data.
- (II.3) **No Grids:** No grids are available, making it harder to read the values.
- (IV.1) **Scales:** The left and right Y-axes have different scales.
- (V.1) **Symbol Definitions:** Assuming "Response time", "Utilization", and "Throughput" are symbols, their definitions should be clear.
- (VII.1) **Last but not Least:** The presence of 3 Y axes is too much, it's not elegant.

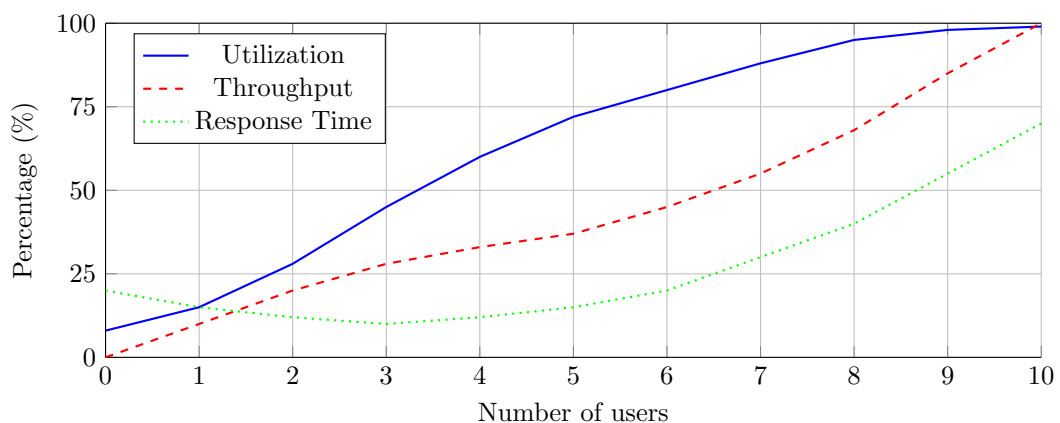


Figure 6: Graph showing the relationship between the number of users and system performance metrics: utilization (blue), throughput (red), and response time (green). As the number of users increases, utilization and throughput both rise, with throughput demonstrating a more exponential growth. Meanwhile, response time increases, particularly after the number of users surpasses a certain threshold.